

Suman Saha

Ph.D. Candidate at UC San Diego

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Education

University of California San DiegoPh.D. in Chemistry | Supervisor: **Prof. Francesco Paesani**

Expected graduation year: 2028

GPA: 3.9/4.0**Indian Institute of Technology Kanpur**

M.S. in Chemistry

Jul. 2021 - Mar. 2023

CGPA: 9.8/10.0**St. Xavier's College, Kolkata | Calcutta University**

B.S. in Chemistry

Jul. 2018 - Jun. 2021

CGPA: 8.12/10.00

Publications

(* represents corresponding author & † represents equal contribution.)

7. **Suman Saha***, Francesco Paesani*. "Hydration Free Energies of Alkali Metal and Halide Ions from Data-Driven Many-Body Potentials." *J. Chem. Theory Comput.*, 2026, ASAP article. [doi:10.1021/acs.jctc.6c00391](https://doi.org/10.1021/acs.jctc.6c00391).
6. Leonard Dick, Richa Rashmi, Henry Agnew, **Suman Saha**, Xuanyu Zhu, Barbara Kirchner*, Francesco Paesani*. "Carbon Polarization and Carbon Nanotube Stacking Impacts the Behavior of Water in Nanoconfinement." *ACS Nano*, 2026, **20**, 1304–1315. [doi:10.1021/acsnano.5c17458](https://doi.org/10.1021/acsnano.5c17458).
5. Saswata Dasgupta*, **Suman Saha**, Francesco Paesani*. "Reactivity of Nanoconfined Water Is Modulated by the Properties of Confining Materials." *J. Phys. Chem. Lett.*, 2025, **16**(46), 11996–12004. [doi:10.1021/acs.jpcllett.5c02963](https://doi.org/10.1021/acs.jpcllett.5c02963).
4. Saswata Dasgupta*, **Suman Saha**, Francesco Paesani*. "Sub-nanometer Confinement Suppresses Autoionization of Water." *J. Am. Chem. Soc.*, 2025, **147**(29), 25167–25173. [doi:10.1021/jacs.5c08273](https://doi.org/10.1021/jacs.5c08273).
3. Yuanhui Pan†, **Suman Saha†**, Matthew Burigana, Vivek Singh, Omar M. Yaghi, Francesco Paesani*. "Thermodynamics of Alkali Metal Ion Uptake from Aqueous Solution in MOF-808." *Chem. Sci.*, 2025, **16**, 12129–12138. [doi:10.1039/D5SC01596K](https://doi.org/10.1039/D5SC01596K).
2. Etienne Palos, Ethan F. Bull-Vulpe, Xuanyu Zhu, Henry Agnew, Shreya Gupta, **Suman Saha**, Francesco Paesani*. "Current Status of the MB-pol Data-Driven Many-Body Potential for Predictive Simulations of Water Across Different Phases." *J. Chem. Theory Comput.*, 2024, **20**(21), 9269–9289. [doi:10.1021/acs.jctc.4c01005](https://doi.org/10.1021/acs.jctc.4c01005).
1. Vaishali Thakkur, Chandan Kumar Das, Shivani Verma, **Suman Saha**, Nisanth N. Nair*. "Methyl Substituted β -Lactam Framework Based Antibiotics and β -Lactamase Inhibitors: Proof of Concept by Computations." Accepted in *ChemPhysChem*. Preprint: [bioRxiv](https://doi.org/10.26434/chemrxiv-2025-12345).

Research Experience

Hydration Free Energies under Different Confinement Conditions

May. 2026 - Ongoing

Supervisor: **Prof. Francesco Paesani**, UC San Diego

- Investigating how different confinement environments alter hydration free energies using molecular dynamics and free-energy calculations.

Ion Transport Mechanisms in Anionic Metal-Organic Frameworks

Sep. 2025 - Ongoing

Supervisor: **Prof. Francesco Paesani**, UC San Diego

- Investigating ion transport mechanisms in anionic metal-organic frameworks using molecular dynamics and free-energy methods to resolve diffusion pathways and framework-mediated selectivity.

Hydration Free Energies from Data-Driven Many-Body Potentials

Sep. 2025 - Apr. 2026

Supervisor: **Prof. Francesco Paesani**, UC San Diego

- Developed staged alchemical free-energy calculations for alkali metal and halide ions in MB-pol water, including robust soft-core Coulomb scaling in the MBX electrostatics framework.

- Ice-Binding Mechanisms in AI-Designed Antifreeze Proteins** Oct. 2024 - Ongoing
 Supervisor: **Prof. Francesco Paesani**, UC San Diego
- Investigated how AI-designed antifreeze proteins recognize and bind ice surfaces using molecular dynamics simulations to characterize interfacial ordering and adsorption motifs.
- Nanoconfined Water Autoionization and Interfacial Reactivity** Jun. 2024 - Ongoing
 Supervisor: **Prof. Francesco Paesani**, UC San Diego
- Investigated how sub-nanometer confinement, interfacial chemistry, and framework flexibility modulate acid-base equilibria and proton-transfer pathways in water.
- Ion Adsorption in Metal-Organic Frameworks** Jan. 2024 - Feb. 2025
 Supervisor: **Prof. Francesco Paesani**, UC San Diego
- Studied solvated alkali metal ions in MOF-808 using alchemical free-energy calculations and enhanced sampling methods.
- Binding Affinity and Free-Energy Calculations in Biomolecular Systems** Oct. 2022 - Apr. 2023
 Supervisor: **Prof. Nisanth N. Nair**, IIT Kanpur
- Evaluated drug-protein binding interactions through thermodynamic integration and related molecular simulation techniques.

Skills

Programming Languages: Python, Fortran, Bash

Molecular Dynamics Packages: LAMMPS, ASE, CHARMM, AMBER, GROMACS

Quantum Mechanics Packages: CP2K, ORCA, Gaussian

Neural Network Potential Packages: DeepMD-kit, NequIP, MACE, UMA

Other Software: VMD, Avogadro2, GaussView, L^AT_EX, ChemDraw, GNUplot, Origin, Matlab, Mathematica

Teaching:

- Graduate Teaching Assistant** for Gen. Chem. Lectures (CHEM 6B) at **UC San Diego**. Spring 2025
- Graduate Teaching Assistant** for Gen. Chem. Lab (CHEM 7L) at **UC San Diego**. Spring 2024
- Graduate Teaching Assistant** for Gen. Chem. Lab (CHEM 7L) at **UC San Diego**. Winter 2024
- Graduate Teaching Assistant** for Gen. Chem. Lectures (CHEM 6A) at **UC San Diego**. Fall 2023
- Doubt Clearing Educator (Online) at **Unacademy**. Mar. 2021 - May 2022
- Subject Matter Expert (Online) at **Chegg**. Jul. 2021 - Aug. 2023

Conferences and Workshops

- Global Physics Summit 2026** Mar. 2026
 Gave a talk on “*Confinement-Governed Autoionization of Nanolayered Water.*”
- SoCal TheoChem 8 Symposium** Feb. 2026
 Presented a poster on “*Thermodynamics of Alkali Metal Ions in MOF-808.*”
- Chemistry and Physics of Liquids, Gordon Research Conference** Aug. 2025
 Presented a poster on “*Thermodynamics of Alkali Metal Ion Uptake from Aqueous Solution in MOF-808.*”
- Quantum-Mechanical, Data-driven, & Multi-scale Simulation of Complex System Workshop** Jul. 2025
 Presented a hands-on tutorial on molecular simulations using the MB-nrg potential with MBX and LAMMPS.
- American Chemical Society 2025 Meeting** Mar. 2025
 Presented a poster on “*Accurate Modeling of Aqueous Phase Chemistry with Rigorous Physical Principles.*”
- Joint March Meeting and April Meeting: Global Physics Summit 2025** Mar. 2025
 Presented a poster on “*Accurate Computational Modeling of Aqueous Phase Chemistry Across Phases and Interfaces.*”

Scholastic Achievements

- Academic Excellence in the Master of Science** 2022
 Received the **Merit for Academic Excellence in the Master of Science in Chemistry** at IIT Kanpur for the year 2021 - 2022.
- All India Rank - 80, IIT-JAM'21** 2021
 Secured **All India Rank 80** in **IIT-JAM'21** (Joint Admission Test for Masters), Chemistry.
- Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship** 2018 - 2023
 \$5,000 fellowship from the Government of India for outstanding undergraduate students for five years.

Jagadish Bose National Science Talent Search (JBNSTS) Fellowship

2016 - 2018

\$700 fellowship from the Government of West Bengal for excellence in studies in higher education.

Certifications

Machine Learning Specialization

May 2023 - Aug. 2023

An online non-credit specialization authorized by **DeepLearning.AI** and **Stanford University** and offered through Coursera ([Credential Link](#)).

Quantum Mechanics of Molecular Structures

Jan. 2021 - Mar. 2021

A course of study offered by **UTokyoX**, an online learning initiative of the **University of Tokyo** on the edX platform ([Credential Link](#)).

Extracurricular Activities

Mentor, UCSD Mentor Assistance Program (MAP)

Sep. 2024 - Ongoing

Mentor high school students through the [UCSD Mentor Assistance Program \(MAP\)](#) at the San Diego Supercomputer Center.

Curator at Deepstash

Mar. 2021 - Ongoing

Publish small articles on summaries of self-help books in [Deepstash](#). Profile Link: [ssuman24](#).

Social Work

Feb. 2019 - Apr. 2019

Taught underprivileged children for 12 weeks in an evening school under **St. Xavier's School, Kolkata**.

Leadership Role

Jan. 2020

Core Committee Member at National Seminar "**Recent Advances in Chemical and Biological Sciences: International Chapter**" held at St. Xavier's College, Kolkata.